

NAME SURNAME:..... SIGNATURE:..... NUMBER:.....

1. Compute the following limits in **DETAIL** and **WITHOUT USING DERIVATIVE(!)**

$$\bullet \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} =$$

$$\bullet \lim_{(x \rightarrow 1^+)} \frac{|x-1| + \llbracket x-3 \rrbracket}{x+1} =$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2 + 3x - 2}}{3x + 5} =$$

$$\bullet \lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x} =$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{3x - \sqrt{5}}{\sqrt{9x^2 + 3x + 1}} =$$

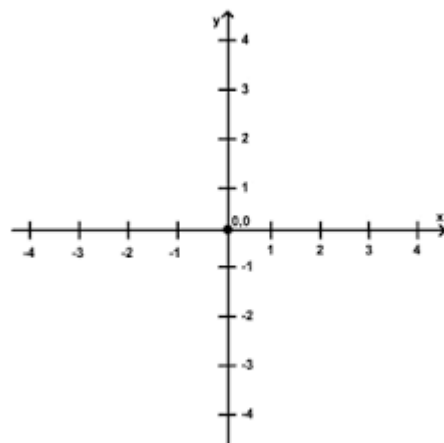
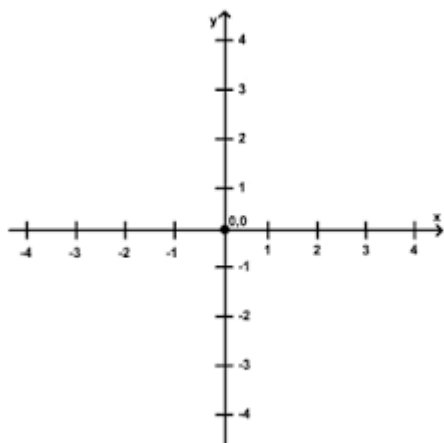
$$\bullet \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + x + 1}}{2x + 5} =$$

$$\bullet \lim_{(x \rightarrow 1^-)} \frac{|x-1| + \llbracket x \rrbracket + 2x}{\sqrt{3}x} =$$

2. Sketch the graphs of the following relations on right coordinate system.

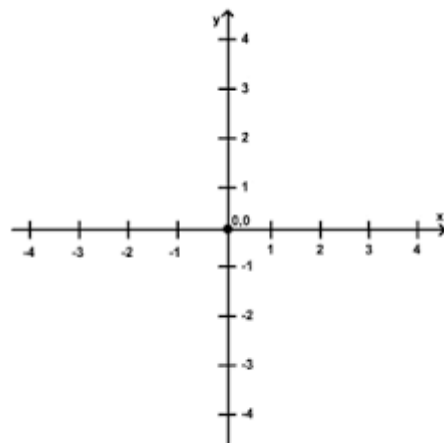
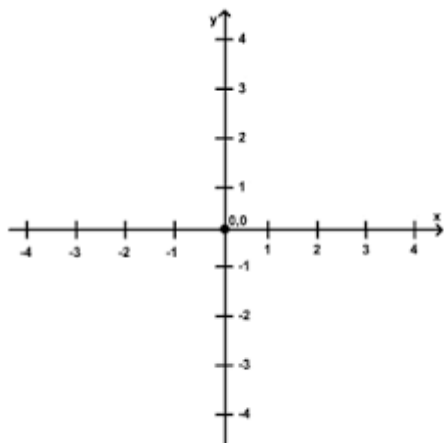
(a) $R = \{(x, y) | (x + 2)^2 + y^2 > 4\}$

(b) $S = \{(x, y) | y > x + 4\}$



(c) $T = \{(x, y) | x^2 + (y - 2)^2 < 3, x \geq 0\}$

(d) $U = \{(x, y) | y \leq |x|\}$



3. Find the solution set of the following inequality analyzing it in two cases ($2x + 3 \neq 0$).

$$\frac{3x - 2}{2x + 3} > 1$$

Case I: $2x + 3 < 0$

Case II: $2x + 3 > 0$